

ParticleNet Architecture

G = batch size (number of events)
 N = number of nodes per event

EdgeConv Block

Node features $(G, N, 9)$

GraphNorm $(G, N, 9) \rightarrow (G, N, 9)$

DynamicEdgeConv ($k=4$) $(G, N, 9) \rightarrow (G, N, 128)$

DynamicEdgeConv ($k=4$) $(G, N, 128) \rightarrow (G, N, 128)$

DynamicEdgeConv ($k=4$) $(G, N, 128) \rightarrow (G, N, 128)$

Aggregation

cat[conv₁, conv₂, conv₃] $(G, N, 384)$

Global mean pool $(G, N, 384) \rightarrow (G, 384)$

Concatenate global features $(G, 384) + (G, 8) \rightarrow (G, 392)$

Graph features $(G, 8)$

Classification

BatchNorm $(G, 392)$

Dense $(G, 392) \rightarrow (G, 128)$

Dense $(G, 128) \rightarrow (G, 128)$

Output $(G, 128) \rightarrow (G, 4)$

$(G, 4)$ raw logits

